Criminal Record Management System

Problem Statement: Manual criminal record keeping is time-consuming, error-prone, and difficult to manage. Searching, updating, and maintaining criminal data using paper files or unorganized systems can lead to data loss, duplication, and inefficiency. Therefore, there is a need for a computerized system that can store, manage, and retrieve criminal records efficiently and securely.

Objectives: This program is a simple console-based system that helps the police keep track of criminal records. It allows authorized users, to enter and view information about criminals. The system is easy to read, organized, and provides a basic level of security.

Tools Used:

1. iostream cin / cout ,
2. string,
3. struct,
4. array,
5. if statements,
6. for loops,
7. login authentication ,
8. manual data entry ,
9. display of records

Functional requirements:

1. User Login
2. Add Criminal Record
3. Store Criminal Data
4. View Criminal Records
5. Limit Record Storage
6. Display Menu
7. Exit System

LANGUAGE USED: C++

SOURCE CODE:

```
#include <iostream>
#include <string>

Using namespace std;

Struct Criminal {
   String firstName;
   String lastName;
   String cnic;
   String crime;
   Int age;
};

Const string adminUser = “admin”;
Const string adminPass = “1234”;

Criminal records[100];
```

```
Int recordCount = 0;

Void policeMenu();
Void citizenMenu();
Void addRecord();
Void viewRecords();
Void searchRecord();
Void editRecord();
Void deleteRecord();
Void generateCertificate();

Int main() {
   Cout << “Criminal Record Management System\n”;

   Int choice;
   Do {
      Cout << “\n1. Police Login”;
      Cout << “\n2. Citizen Check”;
      Cout << “\n3. Exit”;
      Cout << “\nEnter choice: “;
      Cin >> choice;

      If (choice == 1)
         policeMenu();
      else if (choice == 2)
         citizenMenu();
      else if (choice == 3)
         cout << “System Closed.\n”;
      else
         cout << “Invalid choice.\n”;

   } while (choice != 3);

   Return 0;
}

// ---------------- POLICE MENU ----------------

Void policeMenu() {
   String user, pass;
   Cout << “\nUsername: “;
   Cin >> user;
   Cout << “Password: “;
   Cin >> pass;

   If (user != adminUser || pass != adminPass) {
      Cout << “Access Denied!\n”;
      Return;
   }

   Int choice;
   Do {
      Cout << “\n--- Police Menu ---“;
```

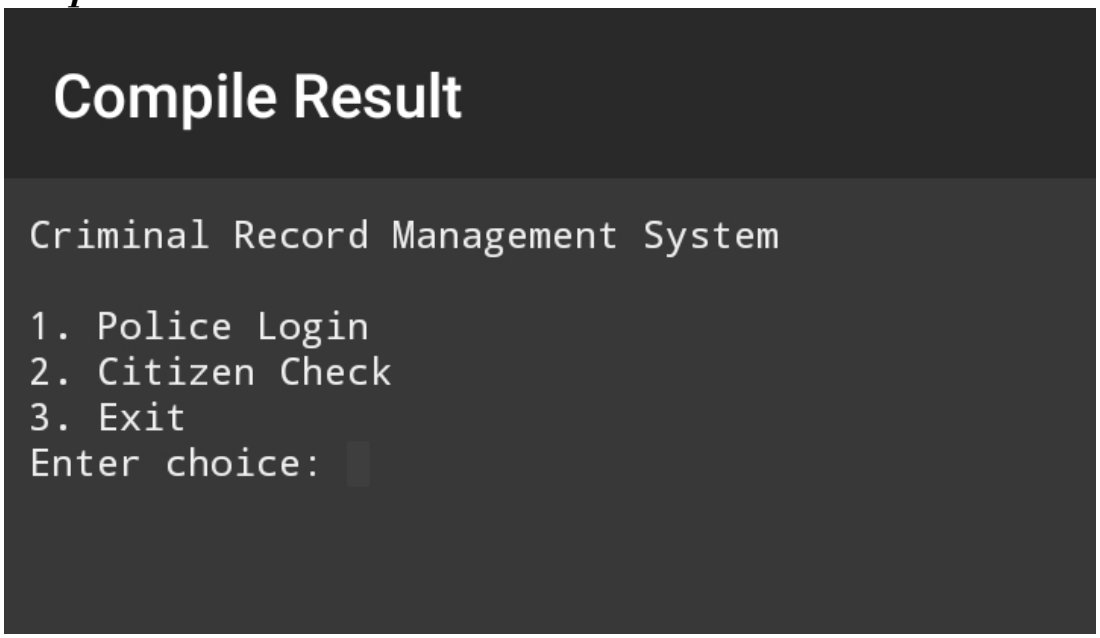
```
        Cout << "\n1. Add Record";
        Cout << "\n2. View Records";
        Cout << "\n3. Search Record";
        Cout << "\n4. Edit Record";
        Cout << "\n5. Delete Record";
        Cout << "\n6. Back";
        Cout << "\nEnter choice: ";
        Cin >> choice;

        Switch (choice) {
        Case 1: addRecord(); break;
        Case 2: viewRecords(); break;
        Case 3: searchRecord(); break;
        Case 4: editRecord(); break;
        Case 5: deleteRecord(); break;
        Case 6: break;
        Default: cout << "Invalid option.\n";
        }
    } while (choice != 6);
}



// ---------------- CITIZEN MENU ----------------

Void citizenMenu() {
    generateCertificate();
}

// ---------------- ADD RECORD ----------------

Void addRecord() {
    If (recordCount >= 100) {
        Cout << "Record limit reached.\n";
        Return;
    }

    Criminal c;

    Cout << "\nEnter First Name: ";
    Cin >> c.firstName;

    Cout << "Enter Last Name: ";
    Cin >> c.lastName;

    Cout << "Enter CNIC: ";
    Cin >> c.cnic;

    Cout << "Enter Age: ";
    Cin >> c.age;

    Cout << "Enter Crime: ";
```

```
    Cin.ignore();
    Getline(cin, c.crime);

    Records[recordCount] = c;
    recordCount++;

    cout << “Record Added (Temporary).\n”;
}

// ---------------- VIEW RECORDS ----------------

Void viewRecords() {
    If (recordCount == 0) {
        Cout << “No records available.\n”;
        Return;
    }
    For (int i = 0; i < recordCount; i++) {
        Cout << “\nName  : “ << records[i].firstName << “ “ << records[i].lastName;
        Cout << “\nCNIC  : “ << records[i].cnic;
        Cout << “\nAge   : “ << records[i].age;
        Cout << “\nCrime : “ << records[i].crime;
        Cout << “\n---------------------------“;
    }
}

// ---------------- SEARCH RECORD ----------------

Void searchRecord() {
    String cnic;
    Cout << “Enter CNIC: “;
    Cin >> cnic;

    For (int i = 0; i < recordCount; i++) {
        If (records[i].cnic == cnic) {
            Cout << “\nRecord Found:”;
            Cout << “\nName  : “ << records[i].firstName << “ “ << records[i].lastName;
            Cout << “\nAge   : “ << records[i].age;
            Cout << “\nCrime : “ << records[i].crime << endl;
            Return;
        }
    }

    Cout << “Record not found.\n”;
}

// ---------------- EDIT RECORD ----------------

Void editRecord() {
    String cnic;
    Cout << “Enter CNIC to edit: “;
    Cin >> cnic;

    For (int i = 0; i < recordCount; i++) {
```

```
        If (records[i].cnic == cnic) {
            Cout << "Enter New First Name: ";
            Cin >> records[i].firstName;

            Cout << "Enter New Last Name: ";
            Cin >> records[i].lastName;

            Cout << "Enter New Age: ";
            Cin >> records[i].age;

            Cout << "Enter New Crime: ";
            Cin.ignore();
            Getline(cin, records[i].crime);

            Cout << "Record Updated.\n";
            Return;
        }
    }

    Cout << "Record Not Found.\n";
}

// ---------------- DELETE RECORD ----------------
Void deleteRecord() {
    String cnic;
    Cout << "Enter CNIC to delete: ";
    Cin >> cnic;

    For (int i = 0; i < recordCount; i++) {
        If (records[i].cnic == cnic) {
            For (int j = i; j < recordCount – 1; j++) {
                Records[j] = records[j + 1];
            }
            recordCount--;
            cout << "Record Deleted.\n";
            return;
        }
    }

    Cout << "Record Not Found.\n";
}

// ---------------- CERTIFICATE ----------------

Void generateCertificate() {
    String cnic;
    Cout << "Enter CNIC: ";
    Cin >> cnic;

    For (int i = 0; i < recordCount; i++) {
        If (records[i].cnic == cnic) {
            Cout << "\n--- CRIMINAL RECORD CERTIFICATE ---";
            Cout << "\nName  : " << records[i].firstName << " " << records[i].lastName;
```

```
        Cout << "\nCNIC  : " << records[i].cnic;
        Cout << "\nAge   : " << records[i].age;
        Cout << "\nCrime : " << records[i].crime;
        Cout << "\n----------------------------------\n";
        Return;
      }
    }

    Cout << "\nStatus: No Criminal Record Found\n";
}
```

Output:



Compile Result

```
Criminal Record Management System

1. Police Login
2. Citizen Check
3. Exit
Enter choice: 1

Username: admin
Password: 1234

--- Police Menu ---
1. Add Record
2. View Records
3. Search Record
4. Edit Record
5. Delete Record
6. Back
```

Compile Result

```
Criminal Record Management System

1. Police Login
2. Citizen Check
3. Exit
Enter choice: 2
Enter CNIC: 41394-3938393-6

Status: No Criminal Record Found

1. Police Login
2. Citizen Check
3. Exit
Enter choice:
```